

Solution

ELECTRICITY -05

Class 10 - Science

1. Suppose we replace the parallel resistors R_1 and R_2 by an equivalent resistor of resistance, R' . Similarly we replace the parallel resistors R_3 , R_4 and R_5 by an equivalent single resistor of resistance R'' . Then using eq. $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$, we have

$$\text{Similarly, } \frac{1}{R''} = \frac{1}{30} + \frac{1}{20} + \frac{1}{60} = \frac{6}{60}$$

that is $R'' = 10\Omega$

Thus, the total resistance, $R = R' + R'' = 18\Omega$

To calculate the current, we use Ohm's law, and get

$$I = \frac{V}{R} = \frac{12V}{18\Omega} = 0.67A$$

2. Resistors 3Ω and 2Ω are in series

The effective resistance of lower arm = $3 + 2 = 5\Omega$

Resistance of upper arm is also 5Ω . The lower and upper arms are in parallel to each other.

Therefore, the total resistance is given by

$$\frac{1}{R} = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}, R_t = \frac{5}{2} = 2.5\Omega \dots\dots\dots (i)$$

The current through ammeter is equal to the total current.

$$\therefore \text{Total current} = \frac{V}{R_t} = \frac{4}{2.5} = \frac{40}{25} = \frac{8}{5} = 1.6A$$

3. a. least count of Ammeter = $\frac{100}{10} = 10\text{ mA}$

$$\text{least count of Voltmeter} = \frac{1}{10} = 0.1V$$

$$\text{b. Voltage, } V = 2 + 0.1 \times 4 = 2.4V \text{ and Current, } I = 200 + 10 \times 5 = 250\text{ mA} = 0.25A$$

$$\text{Thus, the resistance of the resistor, } R = \frac{V}{I} = \frac{2.4}{0.25} = 9.6\text{ ohm}$$

4. Given, $P = 1.5\text{ kW} = 1.5 \times 10^3\text{ W}$, $V = 220\text{ V}$

therefore, the current drawn by a heater, $I = \frac{P}{V}$

$$= \frac{1.5 \times 10^3\text{ W}}{220\text{ V}} = 6.8A$$

Now, the current rating of the fuse ($5A$) is less than the current flowing (6.81 amps) in the circuit. This would mean that the fuse will blow (or melt) and the circuit will shut down, not allowing further current to flow.

5. i. $\therefore$ Power of heater,

$$P = \frac{V^2}{R} = \frac{200 \times 200}{80} = 500W$$

$$\text{ii. } \therefore \text{Heat absorbed by water, } H = mC\theta_R$$

$$= 1 \times 4200 \times 40 [\because \theta_R = 60^\circ - 20^\circ = 40^\circ\text{C}, C = 4200\text{J/kg}^\circ\text{C}]$$

$$= 168000J$$

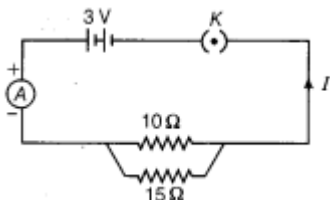
$$= 168kJ$$

$$\text{iii. } H = P \times t$$

$$168000 = 500 \times t$$

$$\Rightarrow t = \frac{168000}{500} = 336s$$

6. Given,



- i. Current through

$$\text{a. } 10\Omega \text{ resistor, } I_1 = \frac{V}{R} = \frac{3}{10} = 0.3A$$

$$\text{b. } 15\Omega \text{ resistor, } I_2 = \frac{V}{R} = \frac{3}{15} = 0.2A$$

- ii. Current flowing through the circuit = $0.3 + 0.2 = 0.5A$

Thus, the ammeter reading is $0.5A$.

$$\begin{aligned}
 7. \quad i. R &= R_1 + R_2 \\
 &= 5 + 10 = 15 \, \Omega \\
 I &= \frac{V}{R} = \frac{2}{15} \\
 &= 0.133 \, \text{A}
 \end{aligned}$$

ii. Potential difference across $10 \, \Omega$

$$\begin{aligned}
 V &= IR \\
 &= \frac{2}{15} \times 10 \\
 &= 1.33 \, \text{V}
 \end{aligned}$$

8. Electric power consumed by

$$\begin{aligned}
 R \text{ is } (P_1) &= I^2 R \\
 2R \text{ is } (P_2) &= I^2 2R \\
 \frac{P_1}{P_2} &= \frac{I^2 R}{I^2 2R} \\
 P_1 : P_2 &= 1:2
 \end{aligned}$$

9. **Ohm's law:** It states that the current flowing through a conductor between two points is directly proportional to the potential difference across the two points.

The arrangement of components in a circuit is normally from the battery to the Ammeter then to the resistor. We have been given the symbols for the Resistor, Ammeter and other components.

The correct order is therefore given as:

B, A, K, R, B

We, therefore conclude that student Y place them in the right order.

10. a. (i) Length of the conductor (l)

(ii) Area of cross-section of the conductor (A)

b. Radius of wire, $r = 0.01 \, \text{cm} = 0.01 \times 10^{-2} \, \text{m}$

Resistance, $R = 10 \, \Omega$

Resistivity, $\rho = 50 \times 10^{-8} \, \Omega \text{m}$

$$R = \rho \frac{l}{A} = \rho \frac{l}{\pi r^2} \Rightarrow l = \frac{R \pi r^2}{\rho}$$

$$\begin{aligned}
 l &= \frac{10 \Omega \times 22 \times (0.01 \times 10^{-2})^2 \, \text{m}}{7 \times 50 \times 10^{-8} \Omega \text{m}} \\
 &= \frac{22}{35} \, \text{m} = 0.629 \, \text{m} = 0.628 \, \text{m} = 0.62 \, \text{m}
 \end{aligned}$$

11. i. Let current through each bulb be I.

$$P = VI, 10 = 220 I$$

$$I = \frac{1}{22} \, \text{A}$$

Let n such bulbs be connected in series.

Current through n bulbs = 5A

n (current in 1 bulb) = 5

$$n \frac{1}{22} = 5$$

$$n = 110$$

110 such bulbs can be lighted within allowable limit of 5A.

ii. Wattage of colour T.V. = 60 W

Number of hours for which colour T.V. is on during September = $2 \times 2\frac{1}{2} \times 30 = 150 \, \text{h}$

$$\text{Energy consumed} = \frac{\text{watt} \times \text{hour}}{1000} = \frac{60 \times 150}{1000} = 9 \text{kWh}$$

Cost of seeing 2 movies per day for 30 days = $9 \times 4 = \text{Rs. } 36$ only

12. a. Here, 2Ω , 3Ω and 6Ω resistance are connected in series.

$$R_{eq} = R_1 + R_2 + R_3$$

$$= 2 + 3 + 6$$

$$= 11$$

$$b. \frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$$

$$\frac{1}{R_{eq}} = \frac{3+2+1}{6}$$

$$\frac{1}{R_{eq}} = \frac{6}{6}$$

$$R_{eq} = 1\Omega$$

13. Potential difference (V) = 9 V.

$$\text{Total resistance (R)} = R_1 + R_2 + R_3 + R_4 + R_5$$

$$= 0.2 + 0.3 + 0.5 + 0.5 + 12 = 13.4\Omega$$

$$\text{Current in the circuit (I)} = \frac{V}{R} = \frac{9V}{13.4\Omega}$$

$$= 0.671 \text{ A.}$$

In series circuit same current flows through all the resistors, hence current of 0.671 A will flow through 12 Ω resistor.

14. i. Zero, current flows due to potential difference and not due to potentials.

$$\text{ii. Energy consumed (in kWh)} = \text{power (in kW)} \times \text{time (h)} = 2.2 \text{ kW} \times 3\text{h} = 6.6 \text{ kWh}$$

$$\text{Power} = 2.2 \text{ kW} = 2.2 \times 1,000 \text{ W} = 2,200 \text{ W}$$

$$\text{But Power} = \text{Voltage} \times \text{Current}$$

$$2,200 = 220 \times I$$

$$I = \frac{2200}{220} = 10 \text{ A}$$

15. R_1 and R_2 are in series,

$$R_s = R_1 + R_2 = 3 + 3 = 6 \Omega$$

R_s and R_3 are in parallel.

$$\therefore \frac{1}{R_p} = \frac{1}{R_s} + \frac{1}{R_3}$$

$$= \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$$\Rightarrow R_p = 2 \Omega$$

R_4 , R_p , R_5 are in series,

$$\therefore R_{s'} = R_4 + R_p + R_5 = 0.5 + 2 + 0.5 = 3 \Omega$$

$$\text{Current, } I = \frac{V}{R_{s'}} = \frac{3}{3} = 1 \text{ A}$$

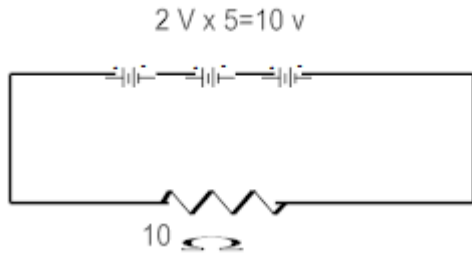
16. Internal resistance, $r = (0.1 + 0.2 + 0.3 + 0.4 + 0.5)\Omega = 1.5 \Omega$

$$\text{Total resistance, } R = (1.5 + 10)\Omega$$

$$= 11.5 \Omega$$

$$I = \frac{V}{R} = \frac{10}{11.5}$$

$$I = 0.869 \text{ A}$$



$$17. \text{ i. Energy consumed} = \frac{\text{Watt} \times \text{hour}}{1000} = \frac{50 \times (2 \times 30)}{1000} = \frac{3000}{1000} = 3 \text{ unit or 3 kWh}$$

$$\text{ii. } R = 20 \Omega; I = 5 \text{ A; } t = 30 \text{ s}$$

$$H = I^2 R t = (5)^2 (20) (30)$$

$$H = 15,000 \text{ J}$$

18. $\therefore$ The resistance of a conductor is directly proportional to the length of the conductor ($R \propto l$).

$$\therefore \text{The resistance of the metallic wire, when it is cut into ten parts of equal length, } r = \frac{R}{10}$$

Two such pieces when joined in series, the equivalent resistance of these two parts = $r + r$

$$= 2r = \frac{2R}{10}$$

$\therefore$ Equivalent resistance of two parts

$$= 2 \times \frac{R}{10}$$

$$= \frac{R}{5}$$

5 such parts are connected in parallel.

$\therefore$ the total resistance R' ,

$$\frac{1}{R'} = \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}}$$

$$= \frac{25}{R}$$

$$\text{Hence, } R' = \frac{R}{25}$$

19. Given, power, $P = 1500\text{W}$, voltage, $V = 230\text{ V}$

i. $\therefore$ Electric down drawn,

$$I = \frac{P}{V} = \frac{1500}{230} = 6\text{A}$$

ii. $\therefore$ Energy consumed, $E = \text{power} \times \text{Time}$

$$= 1500 \times 50$$

$$= 75000\text{Wh}$$

$$= 75\text{kWh} [\because 1\text{ kW} = 1000\text{ W}]$$

$$= 75\text{ unit} [\because 1\text{ unit} = 1\text{ kWh}]$$

$$\text{iii. } \therefore \text{Cost of energy consumed} = 75 \times 6 \\ = \text{Rs } 450.$$

20. The resistance of an ammeter should be low. An ammeter has to be connected in series with the circuit to measure current. In case, its resistance is not very low, its inclusion in the circuit will reduce the current to be measured. In fact, an ideal ammeter is one which has zero resistance.

21. Resistors 1Ω and 2Ω are in parallel.

$$\frac{1}{R} = \frac{1}{1} + \frac{1}{2} = \frac{2+1}{2} = \frac{3}{2}$$

$$R = \frac{2}{3}\Omega$$

$$\text{Total current} = \frac{E}{R} = \frac{2}{\frac{2}{3}} = 3\text{A}$$

The current will divide into two parts in inverse ratio of the resistances.

$$\therefore I_1 = I \frac{2}{1+2} = 3 \times \frac{2}{3} = 2\text{A through } 1\Omega \text{ resistor.}$$

$$\text{and } I_2 = I \frac{1}{1+2} = 3 \times \frac{1}{3} = 1\text{A through } 2\Omega \text{ resistor.}$$

22. a. When an electric current passes through a conductor (like a high resistance wire) the conductor becomes hot after some time and produces heat. This is called heating effect of Electric Current.

Ex: A bulb becomes hot after its use for some time. This is because of heating effect of electric current.

b. We Know that

$$H = P \times t$$

$$H = \frac{\text{Energy}}{\text{Time}} \times t$$

$$\therefore \text{Energy} = V.Q$$

$$H = \frac{V.Q.t}{t}$$

$$H = Vit$$

$$\therefore V = IR$$

$$H = I^2Rt$$

Other form

$$H = V \left[\frac{V}{R} \right] t$$

Where

I = Current

R = Resistance

t = Time

c. Two devices which work on the heating effect of electric current are:

i. Electric heater

ii. Electric iron

23. $R_1 = 5\Omega$, $R_2 = 10\Omega$, and $R_3 = 30\Omega$

Potential difference across the battery, $V = 12\text{ V}$.

This is also the potential difference across each of the individual resistor; therefore, to calculate the current in the resistors, we use Ohm's law.

$$\text{The current } I_1, \text{ through } R_1 = \frac{V}{R_1}$$

$$I_1 = \frac{12\text{V}}{5\Omega} = 2.4\text{ A}$$

$$\text{The Current } I_2, \text{ through } R_2 = \frac{V}{R_2}$$

$$I_2 = \frac{12V}{10\Omega} = 1.2A$$

The current I_3 , through $R_3 = \frac{V}{R_3}$

$$I_3 = \frac{12V}{30\Omega} = 0.4A$$

The total current in the circuit,

$$I = I_1 + I_2 + I_3$$

$$= (2.4 + 1.2 + 0.4)A$$

$$= 4A$$

The total resistance R_p , is given by $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$\frac{1}{R_p} = \frac{1}{5} + \frac{1}{10} + \frac{1}{30} = \frac{1}{3}$$

Thus, $R_p = 3\Omega$

24. a. **Step 1:** Given that:

Length(l) of the metal wire = 2m

Area of cross section (A) = $1.55 \times 10^{-6} \text{m}^2$

Resistivity(ρ) of the metal = $2.8 \times 10^{-8} \Omega\text{-m}$

Step 2: Calculation of resistance:

Resistance of a conductor is given by:

$$R = \frac{\rho l}{A}$$

Thus, putting all the values, we get

$$R = \frac{2.8 \times 10^{-8} \Omega\text{-m} \times 2\text{m}}{1.55 \times 10^{-6} \text{m}^2}$$

$$R = \frac{5.6 \times 10^{-2}}{1.55}$$

$$R = 3.61 \times 10^{-2} \Omega$$

Thus,

The resistance of the metal wire is $3.61 \times 10^{-2} \Omega$

b. Alloys are used in electrical heating devices rather than pure metals because the resistivity of an alloy is more than the resistivity of pure metal. Also, the melting point of an alloy is high, so it does not melt or oxidize easily even at a higher temperature.

25. i. $\rho = R \frac{A}{l}$

$$\text{SI unit of } \rho = \text{ohm} \times \frac{\text{m}^2}{\text{m}}$$

$$= \text{ohm} \times \text{metre}/\Omega \text{ m}$$

ii. $\rho = R \frac{A}{l}$

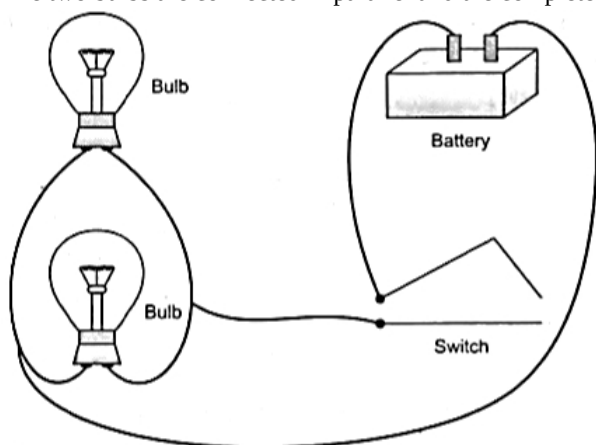
$$= \frac{0.04\Omega \times 1.4 \times 10^{-6} \text{m}^2}{2 \text{m}}$$

$$= 2.8 \times 10^{-8} \Omega\text{m}$$

26. The brightness of the glow of bulb P will increase and brightness of the glow of bulb Q will decrease. This is because on closing S, bulbs Q and R will be in parallel and the combination will be in series with bulb P. Hence the total resistance of the circuit will decrease and the current flowing in the circuit will increase. Therefore, the glow of bulb P will increase.

Also, since bulbs Q and R will be in parallel in this case, the current gets divided and lesser current flows through Q and hence the glow of bulb Q will decrease.

27. The two bulbs are connected in parallel and the complete circuit diagram is drawn below:



The reason for connecting the two bulbs in parallel is that (i) both the bulbs glow at the same voltage, and (ii) if one bulb stops glowing, the other bulb remains unaffected.

28. i. Energy consumed = $P \times t = 2 \times 3 = 6 \text{ kWh}$

Total cost = $6 \times \text{cost per unit}$

Cost per unit = ₹4 per kWh

Therefore, total cost = $6 \times 4 = ₹24$

- ii. We know that,

$$P = \frac{V^2}{R}$$

Thus, resistance is inversely proportional to power. So, the electric lamp with power rating 40 W will have a higher resistance as compared to 60 W lamp.

Lamps with higher power will glow brighter, the lamp with rating 60W, 220V will glow brighter.

29. a. We are given $V = 220 \text{ V}$; $R = 1200 \Omega$

From Eq. ($R = \frac{V}{I}$), we have the current $I = \frac{220V}{1200\Omega} = 0.18A$

- b. We are given, $V = 220 \text{ V}$, $R = 100 \Omega$

From Eq. ($R = \frac{V}{I}$), we have the current $I = \frac{220V}{100\Omega} = 2.2A$

Note the difference of current drawn by an electric bulb and electric heater from the same 220 V source.

30. Current in the circuit = 0.25 A

Current through 4Ω wire = 0.25 A

- a. P.D. across $4\Omega = 0.25 \times 4 = 1V$

- b. P.D. across $3\Omega = 0.25 \times 3 = 0.75 \text{ V}$

31. i. since resistances are in parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{5} + \frac{1}{10} = \frac{8}{10}$$

$$R = \frac{10}{8}\Omega$$

- ii. Total current $I = \frac{V}{R} = \frac{6V}{10/8\Omega}$

$$= 4.8 \text{ A}$$

- iii. If I_1 , I_2 and I_3 be the current through 2Ω , 5Ω and 10Ω respectively.

$$\text{Therefore, } I_1 = \frac{V}{R_1} = \frac{6}{2}$$

$$= 3 \text{ A}$$

$$I_2 = \frac{V}{R_2} = \frac{6}{5}$$

$$= 1.2 \text{ A}$$

$$I_3 = \frac{V}{R_3} = \frac{6}{10}$$

$$= 0.6 \text{ A}$$

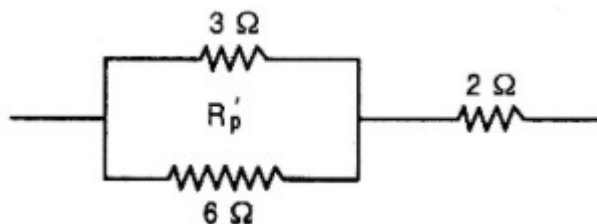
32. $\frac{1}{R_p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = \frac{6}{6} = 1$

Or $R_p = 1\Omega$ when three are in parallel

when 3Ω and 6Ω are in parallel

$$\frac{1}{R'_p} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2}$$

$R'_p = 2\Omega$ is in series with Total 2Ω



$\therefore$ Total resistance when 3Ω and 6Ω are in parallel are in series with 2Ω is $= R'_p + 2 = 2 + 2 = 4\Omega = 4\Omega$

33. i. Zero, current flows due to potential difference and not due to potentials.

- ii. The total energy consumed by the refrigerator in 30 days would be

$$400 \text{ W} \times 8.0 \text{ hour/day} \times 30 \text{ days} = 96000 \text{ Wh}$$

$$= 96 \text{ kWh}$$

Thus the cost of energy to operate the refrigerator for 30 days is

$$96 \text{ kW h} \times ₹ 3.00 \text{ per kWh} = ₹ 288.00$$

34. a. $E = P \times T$

So, $E = 3 \times 2 = 6 \text{ kWh}$

Cost of buying electricity from the main electricity board = $6 \times 3.50 = ₹ 21.0$

b. To reduce the boiling time using the same mains supply, the rate of heat production should be large. We know that $P = \frac{V^2}{R}$.

Since V is constant, R should be decreased. Since R is directly proportional to l so length should be decreased.

35. i. Here $I = 2.5 \text{ mA} = 2.5 \times 10^{-3} \text{ A}$; $V = 12 \text{ volt}$; $R = ?$

$$V = IR \text{ or } R = \frac{V}{I}$$

$$R = \frac{12}{2.5 \times 10^{-3}} = \frac{12}{25 \times 10^{-4}} = \frac{120000}{25} = 4800 \Omega$$

ii. According to the question, $H = 320 \text{ J}$, $t = 10 \text{ s}$, $R = 2 \Omega$, $I = ?$

We know that, $H = I^2 R t$

$$\Rightarrow I = \sqrt{\frac{H}{Rt}}$$

$$= \sqrt{\frac{320}{2 \times 10}}$$

$$= 4 \text{ A}$$

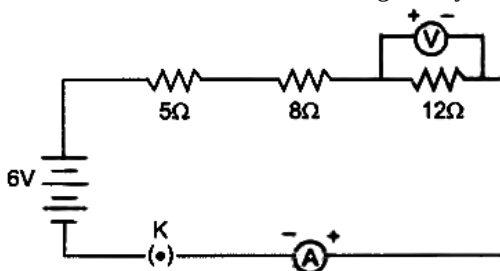
Thus, the amount of current flowing through the resistor is 4 A.

36. a. The devices need to be linked simultaneously. The current flows continuously through the whole electric circuit in a series circuit. Consequently, it is plainly impractical to link an electric heater and light in series since they require currents with quite different values to function. Second, another significant drawback of a series circuit is that when one component fails, the entire circuit is destroyed. A parallel circuit, on the other hand, splits the current among the electrical devices. Thirdly, a parallel circuit has less overall resistance. This is helpful particularly when each gadget has different resistance and requires different current to operate properly.

b. [-] A fuse is connected in series at the beginning of the circuit.

The purpose of a fuse is to prevent over current from damaging the load or other parts of the circuit.

37. The total resistance of the circuit is given by



$$R = 5 + 8 + 12 = 25 \Omega$$

We know, $R = \frac{V}{I}$

Hence, $25 = \frac{6}{I}$

$$I = \frac{6}{25} = 0.24 \text{ A}$$

Since, resistances are connected in series, thus electric current remains the same through all resistors.

Here we have, Electric current, $I = 0.24 \text{ A}$

Resistance, $R = 12 \Omega$

Thus, potential difference (V) through the resistor of 12Ω is given by

$$V = I \times R$$

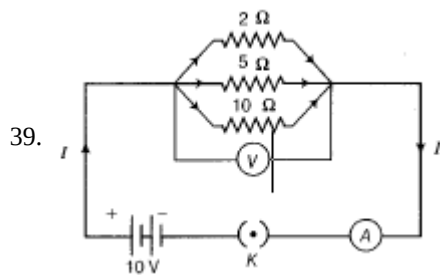
$$= 0.24 \times 12 = 2.88 \text{ V}$$

$\therefore$ Reading of ammeter = 0.24 A

Reading of voltmeter through resistor of $12 \Omega = 2.88 \text{ V}$

38. Advantages of connecting electrical appliances in Parallel with the mains Instead of connecting them in series are

- The voltage across each electrical device are same.
- Separate on/off switches can be applied across each appliances.
- If one electrical device is damaged. It will not affect the other devices.



i. Effective resistance is,

$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{2} + \frac{1}{5} + \frac{1}{10} = \frac{5+2+1}{10} = \frac{8}{10}$$

$$\Rightarrow R_{eff} = \frac{10}{8} = 1.25\Omega$$

ii. Total current,

$$I = \frac{V}{R_{eff}}$$

$$= \frac{10}{1.25}$$

$$= 8A$$

iii. Current through each resistor,

$$I_1 = \frac{V}{R_1} = \frac{10}{2} = 5A,$$

$$I_2 = \frac{V}{R_2} = \frac{10}{5} = 2A$$

and

$$I_3 = \frac{V}{R_3} = \frac{10}{10} = 1A .$$

40. Current in the circuit = 0.25 A

Current through 4Ω wire = 0.25 A

a. P.D. across $R\Omega = 2 - (1 + 0.75) = 0.25$ V

Where R' is the combined resistance of 2 and $R\Omega$ in parallel and is given by

$$\frac{1}{R'} = \frac{1}{R} + \frac{1}{2} = \frac{2+R}{2R} \text{ or } R' = \frac{2R}{2+R}$$

P.D. across $R' = 0.25$ $R' = 0.25 \times \frac{2R}{2+R} = \frac{0.5R}{2+R}$

P.D. across $R' = 0.25$ volt from (c)

$$\therefore 0.25 = \frac{0.5R}{2+R} \text{ or } 1 = \frac{2R}{R+2}$$

$$R + 2 = 2R \text{ Or } r = 2\Omega$$

41. i. $R_{AB} = R_1 + R_2 + R_3$

II. $\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

ii. According to ohm's law ($V = IR$). Slope of V-I graph gives resistance. So greater the slope of graph, greater will be its resistance. Since we know that equivalent resistance in series is greater than the equivalent resistance in Parallel. So the first (I) graph is correct representation of V-I graph in series and Parallel combination.

42. Total resistance

$$R_s = R_1 + R_2 + R_3$$

$$R_s = 5\Omega + 8\Omega + 12\Omega = 25$$

$$V = 6 \text{ volts}$$

$$\therefore I = \frac{V}{R} \text{ (By Ohm's law)}$$

$$I = \frac{6V}{25\Omega} = 0.24 \text{ A}$$

Since all resistances are connected in series therefore current through each resistance will be same. Hence, the reading of ammeter is 0.24 ampere.

Now potential difference across 12Ω resistance

$$V = IR \text{ (By Ohm's law)}$$

$$V = 0.24 \times 12$$

$$V = 2.88 \text{ volts}$$

Since voltmeter connected across 12 ohm resistance, therefore the potential difference across 12 ohm resistance will be the reading of the voltmeter. Hence, the reading of voltmeter is 2.88 volts.

43. Given,

Power $P = 4 \text{ kW}$,

Voltage, $V = 220 \text{ V}$

a. We know that, $P = VI$.

$$\text{Therefore current } I = \frac{P}{V} = \frac{4\text{kW}}{220\text{V}} = \frac{4000\text{W}}{220\text{V}} = 18.18 \text{ A}$$

b. Energy consumed in $2\text{h} = Pt = 4 \text{ kW} \times 2\text{h} = 8 \text{ kWh}$.

c. If cost of $1 \text{ kWh} = ₹ 4.50$, then cost of 8 kWh of energy $= ₹ 4.50 \times 8 = ₹ 36$

44. a. (i) Join the three resistors R_1 , R_2 and R_3 of different values in series connected to a battery of V volt.

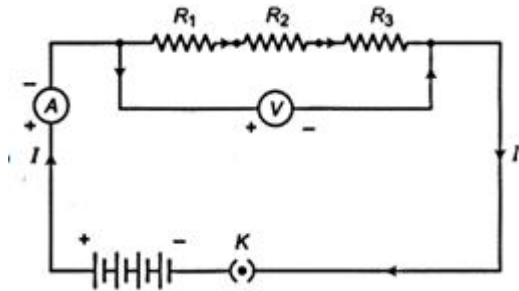
(ii) Connect them with battery, an ammeter (A) and plug key (K).

(iii) Plug the key and note the ammeter reading

(iv) Change the position of the ammeter to anywhere in between the resistors and note the ammeter reading each time.

(v) The ammeter reading will remain the same every time.

Therefore when resistors are connected in series same current flows through all resistors.



b. Given: $R_1 = 5\Omega$, $R_2 = 10\Omega$, $R_3 = 15\Omega$ and $V = 30\text{V}$

Total resistance of the circuit, $R = R_1 + R_2 + R_3$

$$R = 5 + 10 + 15$$

$$R = 30 \Omega$$

Potential difference across the circuit, by ohm's law,

$$V = IR$$

$$\text{or } I = \frac{V}{R} = \frac{30}{30} = 1 \text{ A}$$

$$\text{Potential difference across } 15 \text{ ohm Resistor} = 1 \times 15 = 15 \text{ volt}$$

45. i. $R_s = R_1 + R_2$

$$\text{Here } R_s = 10 \Omega + 50 \Omega = 60 \Omega$$

$$R_3 = 30 \Omega$$

$$\frac{1}{R_p} = \frac{1}{R_s} + \frac{1}{R_3}$$

$$\therefore R = \frac{R_s R_3}{R_s + R_3} = \frac{60\Omega \times 30\Omega}{(60+30)\Omega} = 20\Omega$$

$$\text{ii. } I = \frac{V}{R}$$

$$= \frac{6\text{V}}{20\Omega} = 0.3 \text{ A}$$

46. a. Electric power of an electrical device is defined as its rate of consumption of electrical energy.

$$\text{i.e., } P = \frac{E}{t}$$

The SI unit of electrical power is watt (W)

b. We know,

$$\text{work} = \frac{\text{energy}}{\text{time}}$$

$$\Rightarrow \text{energy} = \text{work} \times \text{time}$$

$$\Rightarrow \text{energy} = 2 \text{ kW} \times 2\text{h} = 4 \text{ kWh}$$

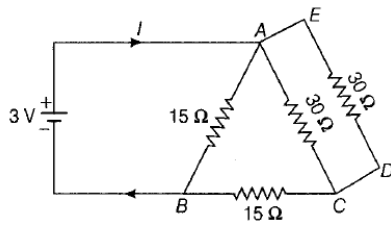
Now,

$$1 \text{ kWh} = 3600000 \text{ joules}$$

$$\Rightarrow 4 \text{ kWh} = 11,400,000 \text{ joules}$$

$$\therefore \text{Energy consumed} = 4 \text{ kWh or } 11,400,000 \text{ joules.}$$

47. According to the question,



i. R_{AC} and R_{ED} are in parallel,

$$\frac{1}{R'_P} = \frac{1}{R_{AC}} + \frac{1}{R_{ED}} = \frac{1}{30} + \frac{1}{30} = \frac{1}{15}$$

$$\Rightarrow R'_P = 15\Omega$$

Now, R'_P and R_{BC} are in series,

$$R'_S = R'_P + R_{BC} = 15 + 15 = 30\Omega$$

Again, R_{AB} and R'_S are in parallel,

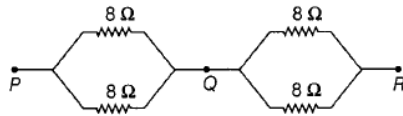
$$\frac{1}{R''_P} = \frac{1}{R_{AB}} + \frac{1}{R'_S} = \frac{1}{15} + \frac{1}{30} = \frac{1}{10}$$

$$\therefore R''_P = 10\Omega$$

So, current is

$$I = \frac{V}{R''_P} = \frac{3}{10} = 0.3A$$

ii. Two 8Ω resistors are connected in parallel. Two such parallel combination must be connected in series to get effective resistance of 8Ω .



48. a. The rate of transfer of electrical energy per unit time is called electrical power.

Its SI unit is Watt.

b. From Joule's law of heating

$$P = \frac{V^2}{R}$$

As bulbs are connected in parallel. So potential difference between them remains same.

$$R_1 = \frac{V^2}{P_1}$$

$$= \frac{(220)^2}{100}$$

$$R_1 = 484\Omega$$

$$R_2 = \frac{V^2}{P_2}$$

$$= \frac{(220)^2}{60}$$

$$R_2 = 806.66\Omega$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{484 \times 806.66}{484 + 806.66}$$

$$R_{eq} = 302.5\Omega$$

Current drawn from the circuit

$$I = \frac{V}{R_{eq}}$$

$$= \frac{220}{302.5}$$

$$= 0.73$$

49. 1. Given, $P = 40 \text{ W}$, $V = 220 \text{ V}$, $I = 6 \text{ A}$

$$\text{We know that, } R = \frac{(V)^2}{P}$$

$$= \frac{(220)^2}{40}$$

$$= 220 \times \frac{220}{40}$$

$$= \frac{48400}{40}$$

$$= 1210\Omega$$

$$\text{Now, } V = IR_{eq}$$

$$\Rightarrow R_{eq} = \frac{V}{I} = \frac{220}{6} = 36.66\Omega$$

Let x bulbs are connected in parallel. So,

$$\frac{1}{36.66} = \frac{x}{1210}$$

$$\therefore x = \frac{1210}{36.66} = 33.066 = 33$$

$$2. \text{ Energy consumed } E_1 = P_1 t_1 = 250 \text{ W} \times 1 \text{ h} = 250 \frac{\text{J}}{\text{S}} \times 3600 \text{ s}$$

$$E_1 = 900000 \text{ J}$$

$$E_2 = P_2 t_2 = 12000 \text{ W} \times 10 \text{ min}$$

$$E_2 = 1200 \frac{\text{J}}{\text{S}} \times 600 \text{ s}$$

$$E_2 = 720000 \text{ J}$$

$\therefore 250\Omega$ TV set consumes more energy.

50. i. Arm AB-Downward, Arm CD-Upward

ii. P and Q-Split ring/Commutator

iii. Arm AB upward, Arm CD downward/Direction of force will get reversed

iv. Fleming's left-hand rule

51. i. Wattage of colour T.V. = 60 W

$$\text{Number of hours for which colour T.V. is on during September} = 2 \times 2\frac{1}{2} \times 30 = 150 \text{ h}$$

$$\text{Energy consumed} = \frac{\text{watt} \times \text{hour}}{1000} = \frac{60 \times 150}{1000} = 9 \text{ kWh}$$

$$\text{Cost of seeing 2 movies per day for 30 days} = 9 \times 4 = ₹36 \text{ only}$$

ii. Energy consumed (in kWh) = power (in kW) $\times$ time (h) = 2.2 kW $\times$ 3h = 6.6 kWh

$$\text{Power} = 2.2 \text{ kW} = 2.2 \times 1,000 \text{ W} = 2,200 \text{ W}$$

$$\text{But Power} = \text{Voltage} \times \text{Current}$$

$$2,200 = 220 \times I$$

$$I = \frac{2200}{220} = 10 \text{ A}$$

52. i. Energy consumed per day by refrigerator

$$= 0.4 \text{ kW} \times 10 \text{ h} [\therefore \text{power of refrigerator} = 400 \text{ W} \frac{400}{1000} \text{ kW} = 0.4 \text{ kW}]$$

$$= 4 \text{ kWh}$$

Energy consumed per day by fans

$$= 2 \times 0.08 \text{ kW} \times 6 \text{ h} [\therefore \text{power of each fan} = \frac{80}{1000} = 0.08 \text{ kW}]$$

$$= 0.96 \text{ kWh}$$

ii. Energy consumed per day by electric tubes

$$= 6 \times 0.018 \text{ kWh} \times 6 \text{ h} [\therefore \text{power of each electric tube} = \frac{18}{1000} = 0.018 \text{ kW}]$$

$$= 0.648 \text{ kWh}$$

$$\text{Total energy consumed per day} = 4 + 0.96 + 0.648 = 5.608 \text{ kWh}$$

$$\text{Energy consumed in 30 days} = 30 \times 5.608 = 168.24 \text{ kWh}$$

Cost of 168.24 kWh

$$\text{Cost of 168.24 units @ ₹ 3.00} = 168.24 \times 3 = ₹ 504.72$$

53. Resistance, 1Ω and 2Ω are in series and combined resistance i.e. $1 + 2 = 3\Omega$ in parallel with 3Ω .

$$\text{Hence total resistance of the combination is } \frac{1}{R} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \text{ or } R = \frac{3}{2} = 1.5\Omega$$

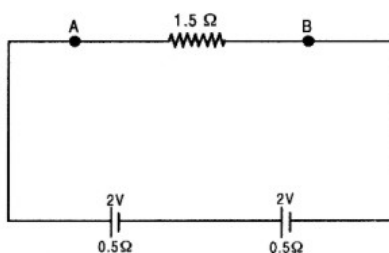
$$\text{i. Total resistance of the circuit} = R + r = 1.5 + 0.5 = 2\Omega$$

$$\text{ii. Total current through ammeter} = \frac{E}{R + r} = \frac{2}{2} = 1 \text{ A}$$

$$\text{iii. In second case total e.m.f.} = 2 + 2 = 4 \text{ V}$$

$$\text{Total resistance} = 1.5 + 0.5 + 0.5 = 2.5\Omega$$

$$\text{Current through circuit in second case} = \frac{4}{2.5} = \frac{40}{25} = \frac{8}{5} = 1.6 \text{ A}$$



54. Given :-

Resistor, $R_1 = 4 \Omega$

Resistor, $R_2 = 8 \Omega$

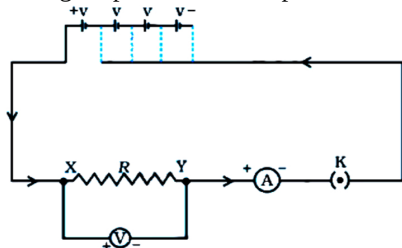
Resistor, $R_3 = 8 \Omega$

Potential Difference, $V = 8$ volts.

i. The potential difference across 4Ω resistors, $V = IR = 1 \times 4 = 4V$

ii. Power dissipated in 4Ω resistors, $P = I^2R = (1A)^2(4\Omega) = 4W$

55. According to Ohm's law, the potential difference (V) across the ends of a resistor is directly proportional to the current (I) flowing through it provided its temperature remains the same.



i.e., $V \propto I$

or $\frac{V}{I} = \text{constant} = R$

or $V = RI$

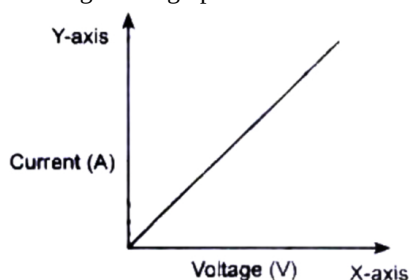
Here R is a constant for the given resistor at a given temperature and is called its resistance.

The SI unit of resistance is ohm (Ω).

A straight line that travels through the origin represents a graph between the potential difference between a resistor's two ends and the current that flows through it. This graph's slope provides the resistor's resistance R .

We test the potential difference across two resistor ends at various currents flowing through it in an electric circuit to confirm Ohm's law. An ammeter is connected in series with the resistor to measure the current flowing through it. By placing a voltmeter in parallel with the resistor, the potential difference between its two ends may be determined.

A straight-line graph obtained between V and I verify the ohm's law.



56. Advantages of connecting electrical devices in parallel with the battery are as follows:

- The voltage across each connecting electrical device is same and the device consumes current as per its resistance.
- Separate on/off switches can be applied across each device.
- Total resistance in parallel circuit decreases, hence, a great current may be drawn from the cell.
- If one electrical device is damaged; then other devices continue to work properly. Moreover, total resistance in parallel circuit arrangement decreases. Hence, there is less heat dissipated as compared to series connection.

57. i. $1 + 2 = 3\Omega$ when connected in series

$$\text{current through each resistor } I = \frac{V}{R_s} = \frac{6}{3} = 2A$$

$$\text{Power used in } 2\Omega \text{ resistor } P_1 = I^2R = (2)^2(2) = 8W$$

$$P_1 = 8W = \frac{V}{2} = \frac{4}{2} = 2A$$

ii. Current through 2Ω resistor

$$\text{Power across } 2\Omega = I^2R = (2)^2(2)$$

$$P_2 = 8W$$

$$\frac{P_1}{P_2} = \frac{8}{8} = \frac{1}{1}$$

58. i. Power = $P = VI = 220 \times 5 = 1,100W$

$$\text{Energy consumed in 2 h } (2 \times 60 \times 60s) E = Pt$$

$$E = (1,100) (2 \times 60 \times 60) = 7,920,000 \text{ J}$$

$$E = 7.92 \times 10^6 \text{ J}$$

ii. Rate at which heat is developed in the heater is its power.

$$P = I^2 R = (15)^2 (8) = 225 \times 8 \text{ or } P = 1800 \text{ W or } \text{Js}^{-1}$$

1,800 J of heat is developed per second.

59. (a) We know that

$$P = \frac{V^2}{R}$$

$$\text{Therefore, } R = \frac{V^2}{P}$$

Resistance of 1st lamp,

$$R_1 = \frac{V^2}{P} = \frac{220 \times 220}{100}$$

$$= 484 \Omega$$

Resistance of 2nd lamp,

$$R_2 = \frac{220 \times 220}{60}$$

$$= \frac{2420}{3} \Omega$$

Since, two lamps are connected in parallel, so its equivalent resistance is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{484} + \frac{3}{2420} = \frac{8}{2420}$$

$$R = \frac{2420}{8} \Omega$$

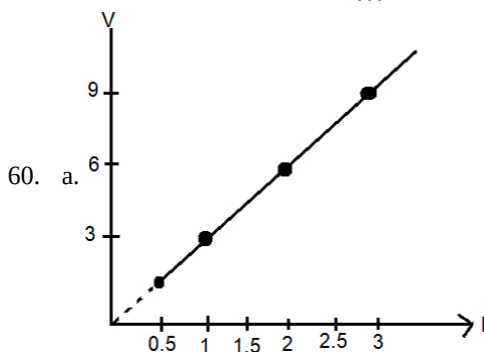
By Ohm's Law, current drawn from the mains:

$$I = \frac{V}{R} = \frac{220 \times 8}{2420}$$

$$= 0.73 \text{ A}$$

$\therefore$ The current drawn from the mains is 0.73 A

$$(b) \text{ Energy consumed} = \frac{\text{Watt} \times \text{hour}}{1000} = \frac{50 \times (2 \times 30)}{1000} = \frac{3000}{1000} = 3 \text{ unit or } 3 \text{ kWh}$$



b. According to ohm's law ($V = IR$). As Current (I) directly, proportional to the voltage. So, As voltage reaches to zero value current in the circuit also become zero or if we extend the line joining coordinates of voltage and current. It will pass through origin.

c. Ohm's law, It states that at constant temperature current through a conductor between two points is directly proportional to the voltage across the two points.

61. Here $R_1 = 2\Omega$; $R_2 = 6\Omega$

Combined resistance R_s when connected in series is

$$R_s = R_1 + R_2 = 2 + 6 = 8 \text{ ohm}$$

When connected in parallel, the combined resistance R_p is

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{2} + \frac{1}{6} = \frac{3+1}{6} = \frac{4}{6} = \frac{2}{3}$$

$$R_p = 1.5 \text{ Ohm}$$

62. a. In first case : $1 = 1\Omega$ and $2 = 10^6\Omega$

$$\therefore \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{1} + \frac{1}{10^6} = \frac{10^6 + 1}{10^6}$$

$$\frac{1}{R_p} = \frac{1,000,000+1}{1,000,000} = \frac{1,000,001}{1,000,000}$$

$$R_p = \frac{1,000,000}{1,000,001} = 0.999999\Omega$$

$$\text{or } R_p = 1\Omega$$

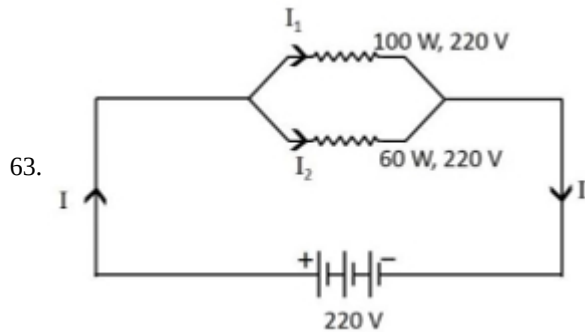
b. In second case, $R_1 = 1\Omega$, $R_2 = 10^3\Omega$, $R_3 = 10^6\Omega$

$$\frac{1}{R_p} = \frac{1}{1} + \frac{1}{10^3} + \frac{1}{10^6} = \frac{10^5 + 10^3 + 1}{10^6}$$

$$\frac{1}{R_p} = \frac{1000000 + 1000 + 1}{1000000} = \frac{1001001}{1000000}$$

$$R_p = \frac{1,000,000}{1,001,001} = 0.999000001\Omega$$

$$R_p = 1\Omega$$



Current drawn by Lamp rated 100 W at 220 V

$$I_1 = \frac{P_1}{V} = \frac{100}{220} = \frac{5}{11} \text{ A}$$

Current drawn by Lamp rated 60 W at 220 V

$$I_2 = \frac{P_2}{V} = \frac{60}{220} = \frac{3}{11} \text{ A}$$

Total current drawn by two lamps in parallel arrangement

$$I = I_1 + I_2$$

$$I = \frac{5}{11} + \frac{3}{11} = \frac{8}{11} \text{ A} = 0.73 \text{ A}$$

64. Rate at which electrical energy is dissipated or consumed in an electric circuit.

Power $P = VI$

$$= \frac{V^2}{R} \quad (I = \frac{V}{R})$$

$$P = \frac{V^2}{R} = \frac{(200V)^2}{400\Omega} = 100W$$

65. The resistance of electric lamp, $R_1 = 20\Omega$,

The resistance of the conductor connected in series, $R_2 = 4\Omega$

Then the total resistance in the circuit

$$R = R_1$$

$$+ R_2$$

$$R_s = 20\Omega + 4\Omega = 24\Omega$$

The total potential difference across the two terminals of the battery

$$V = 6 \text{ V.}$$

Now by Ohm's law, the current through the circuit is given by

$$I = \frac{V}{R}$$

$$\frac{6V}{24\Omega}$$

$$= 0.25A$$

Applying Ohm's law to the electric lamp and conductor separately,

We get potential difference across the electric lamp,

$$V_1 = 20\Omega \times 0.25A$$

$$= 5V$$

and,

$$\text{that across the conductor, } V_2 = 4\Omega \times 0.25A$$

$$= 1V$$

and,

$$\text{that across the conductor, } V_2 = 4\Omega \times 0.25A$$

$$= 1V$$

Suppose that we like to replace the series combination of electric lamp and conductor by a single and equivalent resistor. Its resistance must be such that a potential difference of 6 V across the battery terminals will cause a current of 0.25 A in the circuit.

The resistance R of this equivalent resistor would be

$$R = \frac{V}{I}$$

$$= \frac{6V}{0.25A}$$

$$= 24\Omega$$

This is the total resistance of the series circuit; it is equal to the sum of the two resistances.

66. i. Let current through each bulb be I .

$$P = VI, 10 = 220 I$$

$$I = \frac{1}{22} A$$

Let n such bulbs be connected in series.

$$\text{Current through } n \text{ bulbs} = 5A$$

$$n(\text{current in 1 bulb}) = 5$$

$$n \frac{1}{22} = 5$$

$$n = 110$$

110 such bulbs can be lighted within allowable limit of 5A.

ii. $\therefore$ Power of heater, $P = \frac{V^2}{R} = \frac{200 \times 200}{80} = 500W$

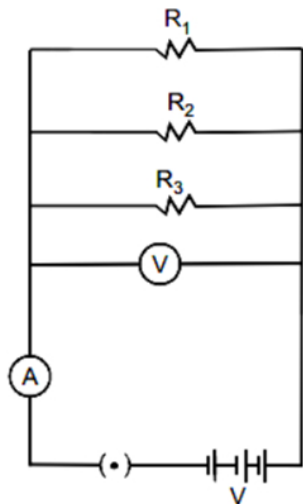
$$\therefore \text{Heat absorbed by water, } H = mC\theta_R$$

$$= 1 \times 4200 \times 40 [\because \theta_R = 60^\circ - 20^\circ = 40^\circ C, C = 4200 J/kg^\circ C]$$

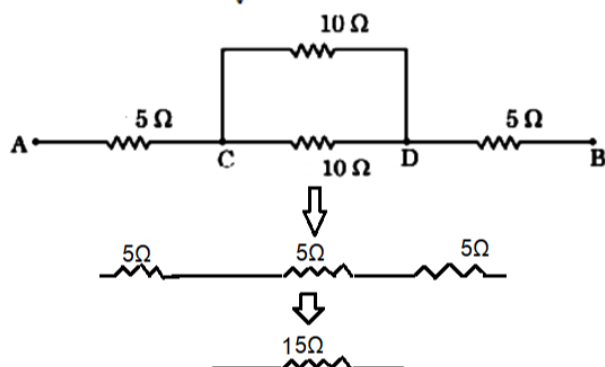
$$= 168000 J$$

$$= 168kJ$$

67. i.



- ii.



Hence, the equivalent Resistance of above network is 15Ω

68. i. Since, all three conductor are made up of same material therefore their resistivity will be same

$$R_A = \frac{\rho L}{A}$$

$$R_B = \frac{\rho \left(\frac{L}{2} \right)}{\frac{A}{2}}$$

$$R_C = \frac{\rho \left(\frac{L}{2} \right)}{\frac{A}{2}}$$

$$a. \frac{R_A}{R_B} = \frac{\frac{\rho L}{A}}{\frac{\rho \left(\frac{L}{2} \right)}{\frac{A}{2}}}$$

$$= \frac{\rho L(2A)}{\rho L(A)}$$

$$= \frac{4}{1}$$

$$\frac{R_A}{R_B} = 4$$

b. $\frac{R_n}{R_c} = \frac{\frac{\rho L}{A}}{\frac{\rho(\frac{L}{2})}{\frac{A}{2}}}$

$$\frac{\rho L \times 2 \times A}{A \times \rho L \times 2}$$

$$\frac{R_A}{R_C} = 1$$

ii. Conductor 'C' has more electrical resistance as its resistivity is more than conductor 'A'.

69. (i) Since $R \propto l$ (length of the conductor)

Length of wire C is more than A and B. Therefore wire C has higher resistance.

(ii) Resistivity of all wires is same as material of all the wires is same. Electrical Resistivity of a wire depends on the nature of the material and not on the dimensions of a wire.



If we connected two 9 ohm resistors in parallel and other 9 ohm resistor in series as shown in figure:

$$\frac{1}{R_p} = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

$$R_p = \frac{9}{2} \Omega$$

$$R = 9\Omega + \frac{9}{2}\Omega = 13.5\Omega$$



If we connected two 9 ohm resistors in series and other 9 ohm resistor in parallel as shown in figure:

$$R_s = 9\Omega + 9\Omega = 18\Omega$$

$$\frac{1}{R} = \frac{1}{18} + \frac{1}{9} = \frac{3}{18}$$

$$\therefore R = 6\Omega$$

71. i. Combined resistance of 100Ω , 50Ω and 500Ω in parallel

$$\text{i.e. } R_p \text{ is given by } \frac{1}{R_p} = \frac{1}{100} + \frac{1}{50} + \frac{1}{500} = \frac{5+10+1}{500} = \frac{16}{500} = \frac{4}{125}$$

Resistance of electric iron = 31.25Ω

$$\text{Current through electric iron} = \frac{V}{R} = \frac{220}{31.25} = 7.04A$$

ii. Energy consumed $E_1 = P_1 t_1 = 250 \text{ W} \times 1\text{h} = 250 \frac{\text{J}}{\text{s}} \times 3600\text{s}$

$$E_1 = 900000\text{J}$$

$$E_2 = P_2 t_2 = 1200\text{W} \times 10 \text{ min}$$

$$E_2 = 1200 \frac{\text{J}}{\text{s}} \times 600\text{s}$$

$$E_2 = 720000\text{J}$$

$\therefore 250\Omega$ TV set consumes more energy.

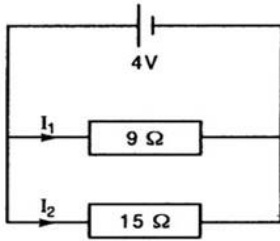
72. Resistor of 6Ω and 3Ω are in series therefore, combined resistance of these two is

$$R_1 = 6 + 3 = 9\Omega$$

Again 12W and 3W are in series. The combined resistance of these two is

$$R_2 = 12 + 3 = 15\Omega$$

Now 9Ω and 15Ω are in parallel to each other. Fig. (a) is the equivalent circuit.



Therefore, the combined resistance of full circuit is

$$\frac{1}{R} = \frac{1}{9} + \frac{1}{15} = \frac{5+3}{45} = \frac{8}{45} \text{ or } R = \frac{45}{8} = 5.6\Omega \dots\dots\dots (i)$$

$V = 4$ volt

$\therefore$ P.D. across 9Ω as well as across 15Ω is 4 V

Current along 9Ω resistor (and hence through 6Ω resistor)

$$I_1 = \frac{V}{R_1} = \frac{4}{9} \text{ A} = 0.44 \text{ A} \dots\dots\dots (ii)$$

Current along 15Ω resistor (and hence through 12Ω resistor)

$$I_2 = \frac{V}{R_2} = \frac{4}{15} \text{ A}$$

Potential difference across 12Ω wire $= 12I_2$

$$= 12 \times \frac{4}{15} = \frac{16}{5} = 3.2V \dots\dots\dots (iii)$$

73. a. **Statement of Joule's law**

Joule's law of heating states that, when a current 'i' passes through a conductor of resistance 'R' for time 't' then the heat developed in the conductor is equal to the product of the square of the current, the resistance and time.

Mathematical explanation

Let H be the heat produced when a current 'i' passes through a conductor of resistance 'r' for time 't' then

i. $H \propto i^2$

ii. $H \propto R$

iii. $H \propto t$

$$\therefore H \propto i^2 R t$$

$$\Rightarrow H = i^2 R t$$

b. Here $V = 6V$ and $R = 5\Omega$

$$\text{The current flowing through the resistor } I = \frac{V}{R} = \frac{6V}{5\Omega} = 1.2 \text{ A}$$

$$\therefore H = I^2 R t = (1.2)^2 \times 5 \times 10 = 72J$$

74. Let the current through the circuit be I which is divided into I_1 and I_2 in the arms AB and CD respectively, then we have

$$I = I_1 + I_2$$

In the arm AB, the total resistance is

$$R_1 = 6\Omega + 3\Omega = 9\Omega$$

and the total resistance in the arm CD is

$$R_2 = 12\Omega + 3\Omega = 15\Omega$$

i. Then current in the 6Ω resistor i.e.,

$$\begin{aligned} I_1 &= \frac{V}{R_1} \\ &= \frac{4}{9} \\ &= 0.44 \text{ A} \end{aligned}$$

ii. Now the current through CD is

$$\begin{aligned} I_2 &= \frac{V}{R_2} \\ &= \frac{4}{15} \\ &= 0.27 \text{ A} \end{aligned}$$

The potential difference across 12Ω , resistor is

$$V_1 = I_2 \times 12\Omega$$

$$V_1 = 0.27 \times 12$$

$$= 3.23 \text{ V}$$